

# Lakshin Ganesha

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## EDUCATION

### University of North Carolina at Chapel Hill

Chapel Hill, NC

Majors: Computer Science (B.S), Statistics and Analytics (B.S) – GPA: 3.8/4.0

Expected – May 2028

- **Organizations:** Member of Edge Carolina, Game Dev, AI@UNC, Analytics and Data Sci, Crypto Investment Club.
- **Relevant Coursework:** Data Structures & Algorithms, Artificial Intelligence, System Programming Fundamentals, Software Engineering Lab, Methods Data Analysis, Probability for Data Science, Computer Vision, Assembly and C, Calculus 3.

## TECHNICAL SKILLS

**Languages:** Python, Java, C, C++, TypeScript, Go, Bash, SQL, HTML, R, C#, Assembly (RISC-V)

**Libraries/Frameworks:** PyTorch, NumPy, FastAPI, Next.js, React, Pandas, Pydantic, Celery, Streamlit, gRPC

**Tools & Skills:** AWS, GCP, Docker, Kubernetes/k8s, Terraform, PostgreSQL, MySQL, CI/CD, Machine Learning, Computer Vision

## EXPERIENCE

### ViiV Healthcare (GSK)

Durham, NC

Data Analytics & AI Co-Op – Incoming

September 2026 – April 2027

### Venuehorn

Apex, NC

Full Stack Software Engineering Intern | FastAPI, PostgreSQL, pgvector, Alembic, Next.js, Pydantic

June 2026 – August 2026

- Developed scalable **RESTful APIs** using **Celery** workers to **decouple long background tasks** from the main event loop.
- Enforced comprehensive data integrity by utilizing **Pydantic** validation layers, **reducing ingestion errors by 40%**.
- Built a **computer vision event ingestion API** using **YOLOv8**, **NumPy**, and semantic tag mapping for venue features.
- Designed a **Next.js** frontend optimizing rendering for a simulated database of **5,000+ users** with strict **JWT authentication**.
- Managed scalable **PostgreSQL** databases utilizing **Alembic** migrations and advanced **pgvector** indexing schemas.

### UNC-Chapel Hill Boerwinkle Lab for Neuroinformatics

Chapel Hill, NC

Machine Learning Engineer & Researcher | LangChain, ChromaDB, Pandas, Pydantic, Ollama

January 2026 – June 2026

- Developed a zero-shot, **automated clinical data extraction pipeline** utilizing a **locally deployed Qwen2.5:32B LLM**.
- Parsed unstructured neurological clinical notes using PHI de-identification mechanisms to guarantee **HIPAA compliance**.
- Implemented **data filtering pipelines** to mitigate LLM hallucinations and accurately capture patient seizure histories.
- Built a validation engine using **Pydantic** and **Pandas**, achieving **96.3% extraction accuracy** in an **FHIR JSON** format.
- Integrated ChromaDB to enable a **natural language Q&A feature**, allowing doctors to retrieve specific medical history.

### Advanced Computational Medicine (ACM)

Chapel Hill, NC

Agentic AI Development Intern | RAG, TrxGemma-27B, LangChain, Vertex AI, Pandas

June 2025 – November 2025

- Collaborated to build a **multi-agent RAG pipeline** to be utilized by doctors and patients for cyber neuro analysis.
- Programmed a **Data Manipulator Agent** using **Pandas** to instantly ingest complex clinical data and resolve schema conflicts.
- Automated **neuro-anomaly detection** by engineering **statistical scripts** to benchmark extracted patient health values.
- Built an **Agent** to synthesize insights from **5,000+ PubMed articles**, achieving **95.8% accuracy** across **500** evaluations.
- Engineered clinical data pipelines achieving **95.8% decision accuracy**, reducing average query responses to **under 20 seconds**.

## PROJECTS

### Quantitative Stock Evaluation Engine | Next.js, TypeScript, FastAPI, PyTorch, Scikit-learn

May 2024 – June 2024

- Created a **full-stack financial platform** using **Next.js** and **FastAPI**, featuring live heatmaps for **500+ S&P equities**.
- Implemented a complex sentiment analysis pipeline with **PyTorch** and **FinBERT** to carefully evaluate daily market sentiment.
- Augmented **Scikit-Learn Random Forest** models to generate accurate, weighted daily price forecasts and confidence scores.
- Developed a portfolio optimizer using **Pandas**, executing **6,500 Monte Carlo simulations** to maximize **Sharpe Ratios**.

### Cloud-Native AI Research Agent | Terraform, AWS (EC2/ECR), Docker, GitHub Actions

February 2025 – April 2025

- Developed a containerized **FastAPI** backend for a **LangChain** and **ChromaDB** powered **RAG pipeline**.
- Deployed an **LLM-based Critic Agent** to automate the accurate detection of data contradictions in dense scientific literature.
- Optimized overall cloud compute efficiency and **reduced latency by 40%** utilizing advanced **HuggingFace** embedding models.
- Built scalable **AWS infrastructure** as code via **Terraform** and **GitHub Actions** for zero-downtime **Docker** deployments.

### NovaKV: Distributed Key-Value Database | Go, gRPC, BoltDB, Protobuf, Raft Consensus

March 2026 – June 2026

- Created a **key-value database** from scratch, using **sync.RWMutex** and **Goroutines** for **sub-millisecond operations**.
- Integrated the **HashiCorp Raft consensus** algorithm to enable automated leader election and continuous fault tolerance.
- Synchronously replicated data across a **multi-node cluster**, ensuring **zero data loss** during severe server failures.
- Implemented **gRPC**, **BoltDB**, and **WAL snapshot compaction** to guarantee data persistence and accelerate crash recovery.